Toulouse School of Economics, 2014-2015
Macroeconomics 1 – Franck Portier

Problem set 1: The Ramsey model

Problem I – The Golden Rule *(easy, do it yourself)*

We consider an economy with the production function

$$Y_t = F(K_t, N_t),$$

where N_t =employment = population and $dN/dt = nN_t$.

The capital accumulation equation is

$$\dot{K}_t = Y_t - C_t - \delta K_t.$$

1 – Rewrite these equations in per-capita term.

2 – Characterize a balanced growth path.

3 – What is the ratio K/N which maximizes consumption per capita? This is called the "Golden rule" level.

4 – Compare the net marginal product of capital with the population growth rate at the golden rule.

5 – Assume factors are paid their marginal product. Compare the share of profits in total income to the share of investment in output at the golden rule.

6 – Assume 'society" maximizes

$$V = \int_0^{+\infty} e^{-\rho t} N_t^{\alpha} \frac{\left(\frac{C_t}{N_t}\right)^{\beta} - 1}{\beta} dt,$$

How would you interpret such a utility function? Show that for the problem to be meaningful we need $\alpha n < \rho$. What happens if this is not the case?

7 – Compare the long-run K/N level to the Golden rule in this case.

8 – How does α affect the long-term stock of capital? Explain.

Problem II – The open-economy Ramsey model with installation costs for capital

We consider an economy which produces a single homogeneous good and can borrow and lend at a fixed rate r (in terms of this good which we will take as the numéraire) from the rest of the world. Its total labor force is normalized to 1 so that its production function is

$$Y_t = AK_t^{\alpha}.$$

The representative consumer's utility function is

$$V = \int_0^{+\infty} e^{-\rho t} C_t^{\beta} dt. \tag{1}$$

We assume that the subjective discount rate is equal to the world interest rate:

$$r = \rho$$

The equation for accumulating capital is

$$\dot{K}_t = I_t, \tag{2}$$

where I_t = investment. To install I units of capital we need to spend $f(I)$ units of the final good, in addition to the I units that are needed for investment. We assume $f(0) = f'(0) = 0$, $f'(x) > 0$ for $x > 0$, $f'(x) < 0$ for $x < 0$, $f''(x) > 0$, $\lim_{x\to\infty} f(x) = \lim_{x\to+\infty} f'(x) = +\infty$,

1 – What is the surplus of the trade balance at an instant t?

2 – Let W_t be the net stock of foreign assets owned by this economy (in terms of the numéraire). What is the evolution of W over time?

3 – Show that if we impose the solvency condition

$$\lim e^{-rt} W_t > 0,$$

then the country has an intertemporal budget constraint given by

$$W_0 + \int_0^{+\infty} e^{-rt}(Y_t - C_t - I_t - f(I_t)) \geq 0.$$

4 – Show that the optimal investment strategy must maximize the following criterion:

$$\tilde{W} = \max \int_0^{+\infty} (Y_t - I_t - f(I_t))e^{-rt} dt$$

subject to (2)

5 – Show that the optimal investment strategy is given by

$$I_t = g(q_t)$$

where $g()$ is a strictly increasing function such that $g(1) = 0$ and q_t satisfies

$$-\dot{q}_t + rq_t = \alpha A K_t^{\alpha-1}$$

6 – What is the economic interpretation of q? Show that it is equal to the present discounted value of future marginal products of capital net of depreciation.

7 – What is the level of capital in steady state? Does the corresponding condition sound familiar?

8 – Draw then phase diagram of this problem. How would you eliminaten the trajectories that do not converge to the steady state?

9 – Using this diagram, describe how investment reacts to a permanent increase in A at date $t = T$, when this increase is anticipated at date $t = 0$. How would you interpret this economically?

10 – Show that the optimal consumption behavior maximizes (1) subject to

$$\int_0^{+\infty} C_t e^{-rt} dt \leq W_0 + W^*,$$

where W^* is the value of $\tilde{W}$ at the optimum investment strategy.

11 – What is the solution to this problem?

12 – Assuming that at $t = 0$ the economy is in steady state with $W_0 = 0$, what is the reaction of the trade balance to the anticipated increase in A discussed in question 9?

Problem III – Decentralization of the equilibrium

We consider the standard Ramsey model, and to simplify we assume no technical progress and no population growth. Thus we have

$$Y_t = F(K_t, L_t), \tag{3}$$

and in equilibrium $L_t = \bar{L} = 1$.

We assume that at any date t there is a continuum of representative firms of mass 1 which all have the same production function and the same capital and employment as in the aggregate economy.

$$\dot{K}_t = Y_t - C_t - \delta K_t.$$

Finally there is a continuum of identical consumers with mass 1, each endowed with L units of labor and their utility function is

$$U = \int_0^{+\infty} u(C_t)e^{-\rho t} dt.$$

At each date there is a competitive market for labor and the wage is w_t. Furthermore, at each date people can borrow and lend at a competitive rate r_t. The supply of loanable funds must be equal to the demand for loanable funds. Finally the firms are owned by the consumers.

1 – Show that the consumers' disposable income is equal to Y_t.

2 – Show that the value of a firm V_t evolves according to

$$r_t V_t = F(K_t, L_t) - w_t L_t - I_t + \dot{V}_t, \tag{4}$$

where I_t are the investment expenditures of the firm at date t. From the point of view of the firm one has $\dot{K}_t = I_t - \delta K_t$, and furthermore it treats L_t not as fixed but as a choice variable.

3 – What is the fundamental value of V_t at t?

4 – Assume that V_t is always equal to its fundamental value and that firms set their optimal investment strategy so as to maximize V_t. Show that at each date t one must have

$$\frac{\partial F}{\partial K}(K_t, L) = r_t + \delta.$$

Interpret.

5 – What is the wage level at t given K_t?

6 – Let W_t be the net stock of assets owned by the representative consumers (that is, the money owed by firms to the consumers at t, expressed in terms of the output good). What is the evolution of W_t over time?

7 – Assume that we impose the solvency condition

$$\lim e^{-rt} W_t > 0,$$

what is the consumer's intertemporal budget constraint?

8 – Solve the consumer's optimization problem and derive the Euler equation.

9 – Show that in equilibrium the Euler equation is the same as on the optimal trajectory and therefore that the Ramsey optimum is also a competitive equilibrium.

10 – Show that $V_t = K_t$. Explain. (Hint: manipulate (4) using the equilibrium factor prices and the fact that $F(,)$ is homogeneous of degree one).